Lower body specialization — Session C (Friday)

Mixed + posterior chain detail · hypermobility-safe · knee rehab phase · ~55 min · 27 working sets

Muscle group

Quads · Hamstrings · Glutes · Calves

30–60 min before: collagen peptides (10–15g) + vitamin C
Maximizes collagen synthesis in tendons loaded during session

B — Superset 1: Hinge + Quad iso (~8 min · 3 rounds · 90s rest after each pair)

1	Banded RDL (long band under feet) Hamstrings · glutes · 3rd hinge variant of week Band tension peaks at top · different curve from DB RDL · soft knee bend	3 × 10–12 Band · 3s ecc

↕ then immediately

2	Wall sit (isometric · feet wider stance) Quads · slight inner-quad emphasis with wider stance Neutral antagonist for RDL · only deepen if pain-free	3 × 25s Bodyweight

C — Superset 2: Hip extension + Hip flexion (~7 min · 3 rounds · 90s rest after each pair)

3	Hip thrust (bodyweight or light DB · controlled tempo) Glute max · 2nd glute compound exposure of week Stop short of full hip extension · drive through heels · 3s eccentric	3 × 12–15 BW / DB · 3s ecc

↕ then immediately

4	Standing banded hip flexion (knee drive) Hip flexors (iliopsoas · rectus femoris top portion) Band anchored low behind · drive knee up · strict antagonist to hip thrust	3 × 12/leg Band · low anchor

D — Superset 3: Calf + Glute iso (~6 min · 3 rounds · 60s rest after each pair)

5	Standing calf raises (Smith · optional deficit) (3s eccentric · 1s pause at bottom stretch · soft knee) Gastrocnemius (primary) · soleus synergist Optional deficit: plate under balls of feet · heel drop ~3–5cm · 70–80% range	3 × 12–15 Smith

↕ then immediately

6	Banded glute bridge isometric hold Glutes · hamstrings (iso stabilizer) Cross-joint pairing (ankle ↔ hip) · zero overlap with calves	3 × 20s Band

E — Finisher: Soleus + Adductor + Tib ant + Abductor + Proprioception (~10 min · straight sets · 45s rest)

7	Seated calf raise (plate on knees) (3s eccentric · 1s pause at bottom stretch · soft lockout) Soleus — knee-flexed position targets soleus over gastroc	3 × 12–15 Plate on knees
8	Copenhagen plank holds (bench-supported) (top leg on bench · bottom leg suspended · 20–30% effort) Adductors · isometric in lengthened position	2 × 15s/side Bodyweight
9	Tibialis raises (wall-supported) Tibialis anterior · ankle dorsiflexion strength Lean back against wall · lift toes · landing/badminton prep	4 × 15 Bodyweight / band
10	Side-lying banded hip abduction Glute medius · hip abductors Knee stabilizer maintenance	3 × 12/side Band
11	Single-leg stance (slight knee bend, eyes closed if able) Proprioception · ankle + hip stabilizers Critical for hypermobile joints · matches cardio day protocol	2 × 20s/leg Bodyweight

Within 60 min after: protein 30–40g + vitamin C
Supports muscle + tendon remodeling · aim 1.6–2g protein/kg/day total

Every rep, every set — hypermobility + knee rehab rules

- Never lock out knees — soft bend at end range
- No single-leg eccentric loading through knee flexion-extension cycles (rehab phase)
- Skip session if joints feel unusually loose that day — reschedule, don't push
- Stop set early if any joint feels unstable — prioritize joint feel over rep count
- No pain rule — stop immediately if any knee pain or swelling
- Lengthened-position work targets ~70–80% of available range — never end range
- Hip abductor/adductor machine: deferred until knee stabilizes

Session timeline

Superset B (8 min) → Superset C (7 min) → Superset D (6 min) → Finisher E (10 min)
Total working session: ~55 min · 27 working sets · then proceed to post-workout mobility protocol

Working set count — Session C

Hamstrings: 3 sets (banded RDL)
Quads: 3 sets (wall sit iso)
Glutes: 6 sets (thrust 3 + bridge 3)
Hip flexors: 3 sets (knee drive)
Calves: 6 sets (standing 3 + seated 3)
Hip abductors: 3 sets (side-lying)
Adductors: 2 sets (Copenhagen iso)
Tibialis ant: 4 sets

Alternative exercises — if Smith machine occupied

#	Primary exercise	Alternative 1	Alternative 2
5	Standing calf raises (Smith)	DB standing calf raise (hold rack)	Single-leg DB calf raise (hold rack)

v3 — warmup removed (redundant with mobility doc) · alts audit (row 5 BB calf raise → single-leg DB)